Chanakya University

Project for Laboratory Examination 2
Deadline for submission: 2 PM on 14-10-2026

Note:

a) Implement the project below. Evaluation at the time of the examination will be carried out for each student separately.
2. You must write a brief description of your design of the project and add meaningful comments in your code.
3. This implementation will not carry any marks on its own. You will also be asked to modify your own code during the examination to add more features, and those changes will carry full 100% marks . Neither internet access nor AI tools will be allowed during the examination.
4. You must submit the implementation (design explanation+code+test cases) to **Mr. Nikhil S., latest by 2 PM on October 14, 2026**. The method of submission will be as in the previous lab exam project. The second lab exam will be tentatively on **October 16, 2026**. A confirmation regarding the date will be sent in due course of time.

Project on Index Generation

Your task is to write a utility that indexes words in a file. Your program should scan a file looking for words, then print out to a text file, a table of all the words found in *lexicographic order* together with a list of line numbers indicating where the word was found. Implement the program in two ways:

a) using a *Trie* data structure, with the key being a word from the file.
b) using a *binary search tree* (BST) data structure, with the key being a word from the file.
c) using Python dictionaries, with the key being a word from the file.

In all the three cases above, repetition of words is permitted and must be taken care of. Each complete word node could contain a linked-list queue of line numbers. Compare the performance of the three implementations (w.r.t. time) using SIX large input text files containing more than a million (10^6) words, with 1000, 5000, 10000, 50000, 100000, and 500000 distinct words. You must compare the performance for three actions:

a) construction time of the complete index using Trie/BST/dictionary along with linked lists.
b) search time using 100000 random words.
c) printing the index with words in lexicographic order.

Plot the three times using Matplotlib. You may use AI tools for any task in the project.

Do not make any restrictive assumptions regarding the text file of words. All printable ASCII characters must be permitted and they may be compared using the ASCII order. Words in the text file may be assumed to be separated with one or more blanks, line breaks, full stop, comma, colon, or semicolon.

Your utility must accept two arguments in the command line: the input text file name and the output index file name. Implement all the three versions in Python.